

# Xuesong Zeng

xuesong.zeng@csu.edu.cn | GitHub | OpenReview | ORCID

## Education

Central South University

Sep. 2023 - Jun. 2027  
(expected)

B.Eng. in Computer Science and Technology; GPA 90.47/100; rank 22/258 (top 10%). CET-4: 632; CET-6: 599.

## Research Experience

Research Assistant, BD4UI Lab, Central South University

Jun. 2025 - Present

Supervised by Senzhang Wang and Miao Hao. Research in spatio-temporal data mining; first author of ViCO, accepted to CIKM 2026 Full Research Track.

Research Assistant, DILab, USTC

May 2026 - Present

Supervised by Pengkun Wang. Research on LLM reasoning and agentic systems.

Visiting Student, NUS

Jun. 2025 - Jul. 2025

Supervised by Wee-Kek Tan; focused on AIoT full-stack systems.

## Selected Projects

ViCO: Vision-Augmented Time Series Anomaly Detection with Prototype-Guided Coordination 2025 - 2026

**First author.** Accepted to CIKM 2026 Full Research Track. Proposed a visual-temporal coordination framework with prototype retrieval for efficient time-series anomaly detection.

- Decoupled temporal numerical modeling and visual structural perception into complementary branches.
- Compressed visual knowledge into a fixed prototype bank to reduce online inference cost.

Multimodal Detection and Trustworthy Explanation Platform for News

Mar. 2025 - Jun. 2025

**Team leader.** Built a two-stage pipeline for image forensics, tampered-region localization, and evidence-based explanations for multimodal news.

- Fine-tuned Qwen2.5-VL-3B-Instruct with LoRA; image-level accuracy improved from 55.83% to 82.33%.
- Achieved 0.84 IoU / 0.90 F1 for tampered-region localization and 97.5% image-text relation accuracy.

Action-SRaR: Action-conditioned Step-wise Rubric Rewards

Jun. 2026 - Present

Developing structured reasoning actions and rubric rewards for process supervision in RLVR. A controller-executor framework with counterfactual action selection reduces redundant checks and repeated steps.

## Honors and Awards

- National 3rd Prize, 17th Service Outsourcing Innovation and Entrepreneurship Competition, AI Track (2026).
- National 3rd Prize, 16th Service Outsourcing Innovation and Entrepreneurship Competition, Enterprise Track (2025).
- National 3rd Prize, College Students Science and Technology Translation Competition (2023).
- Computer Software Copyright, second responsible person (2026).
- First Prize, Central South China Regional Competition, China Collegiate Computer Design Competition (2025).
- Second Prize, Central China Division, China Collegiate Computer Competition AIGC Innovation Competition (2025).

## Additional Activities

- Viola Section Leader, Central South University Student Symphony Orchestra (Sep. 2023 - Present).
- Member, Apple Lab, Central South University (Sep. 2024 - Present); organized Innosed Minicamp, a student-club hackathon.

## Skills

Python, C++, JavaScript, LaTeX, and presentation design.